

4.8 Consequences of uses of computing

4.8.1 Individual (moral), social (ethical), legal and cultural issues and opportunities

Content	Additional information
Show awareness of current individual (moral), social (ethical), legal and cultural opportunities and risks of computing. Understand that: • developments in computer science and the digital technologies have dramatically altered the shape of communications and information flows in societies, enabling massive transformations in the capacity to: • monitor behaviour • amass and analyse personal information • distribute, publish, communicate and disseminate personal information. • computer scientists and software engineers therefore have power, as well as the responsibilities that go with it, in the algorithms that they devise and the code that they deploy. • software and their algorithms embed moral and cultural values. • the issue of scale, for software the whole world over, creates potential for individual computer scientists and software engineers to produce great good, but with it comes the ability to cause great harm. Be able to discuss the challenges facing legislators in the digital age.	Teachers may wish to employ two very powerful techniques, hypotheticals and case studies, to engage students in the issues. Hypotheticals allow students to isolate quickly important ethical principles in an artificially simplified context. For example, a teacher might ask students to explain and defend how, as a Google project manager, they would evaluate a proposal to bring Google's Street View technology to a remote African village. What questions should be asked? Who should be consulted? What benefits, risks and safeguards considered? What are the trade-offs? Case studies allow students to confront the tricky interplay between the sometimes competing ethical values and principles relevant in real world settings. For example, the Google Street View case might be used to tease out the ethical conflicts between individual and cultural expectations, the principle of informed consent, Street View's value as a service, its potential impact on human perceptions and behaviours, and its commercial value to Google and its shareholders. There are many resources available on the Internet to support teaching of this topic.